

Exact two-sided certificates for maximal quantum Bell violations: an agentic laboratory study of the doubly tilted CHSH family

Team Ninja Turtles, Hack-Nation 2026, Challenge 3 (Agentic Scientific Discovery)

Abstract

We study the doubly tilted CHSH family with correlator coefficients $a = b = [p, 0]$ and $c = [[1, 1], [1, -1]]$. An automated laboratory computed lower bounds on the maximal quantum value $Q(p)$ from explicit strategies and upper bounds from the NPA hierarchy. A verifier certified both bounds in exact rational arithmetic. Q is certified to within $1e-6$ at $p = 0.3$ [C-bell-R4], at $p = 1/2$ [C-bell-R1] and at $p = 131/200$ [C-bell-R6]. At $p = 0.95$ and $p = 0.99$ the explicit strategies violate the classical bound $L = 2+2p$, but the two-sided gap is not closed [C-bell-R2, C-bell-R2-RT2]. At $p = 131/200$, NPA level 1+AB is strictly not tight, while level 2 is [C-kombi1]. No closed formula for $Q(p)$ was established. The preregistered anchor hypothesis H6a failed [C-H6a], and no novelty is claimed [C-H6b].

1 Introduction

Question. How does the maximal quantum value $Q(p)$ behave for the doubly tilted CHSH family with $a = b = [p, 0]$? Do qubits suffice? Where does the NPA level 1+AB cease to be tight? [C-bell-R1, C-bell-R5]

Contribution. The study reports individually certified values and gap bounds. It also lists claims that could not be established.

Results.

- Two-sided certificates for Q at three parameter values: $p = 0.3$ [C-bell-R4], $p = 1/2$ [C-bell-R1] and $p = 131/200$ [C-bell-R6].
- A rigorous quantum violation of L at $p = 0.95$ and $p = 0.99$, without a closed two-sided gap [C-bell-R2, C-bell-R2-RT2].
- Strict non-tightness of NPA level 1+AB at $p = 131/200$ [C-kombi1].
- Verifier lower bounds on the gap between the 1+AB optimum and the best certified strategy at six parameter values [C-kombi2].
- A list of negative results, red-team outcomes and unmet preregistered hypotheses, given in the section on negative results [C-neg1, C-H6a].

2 Setting and certificates

The Bell inequality is given in correlator form with coefficients $a = [p, 0]$, $b = [p, 0]$, $c = [[1, 1], [1, -1]]$ and constant 0 [C-bell-R1]. The classical bound is $L = 2+2p$, and at $p = 1/2$ it is $L = 3$ [C-bell-R2].

Lower bounds on Q come from explicit quantum strategies of dimension d per side, found by see-saw optimisation. The verifier evaluates their Bell value exactly in rational arithmetic [C-bell-R1-RT1]. Upper bounds come from NPA levels 1+AB, 2 and 2+AAB, certified by rational dual certificates [C-bell-R1, C-bell-R2-RT1]. Q counts as proved when the gap between lower and upper bound is below $1e-6$ [C-bell-R1]. Rational feasible

moment matrices give lower bounds on the NPA optimum, and the verifier compares these with the best exact strategy [C-bell-R5].

3 Method: the agentic laboratory

The laboratory ran 8 rounds, with 7 checked statements and 1 negative result, at a cost of 4.31 USD [C-methode]. Each round was preregistered before the experiment [C-methode]. Researcher agents proposed claims, and a code verifier certified or rejected them. A red team attacked each claim with counter-checks. Agent interpretations are labelled as unverified hypotheses and are not used here [C-bell-R1-I].

Before the run, the verifier passed a self-test with 22/22 cases correct (10 true, 12 false statements or rule violations) [C-selbsttest]. The anchors were CHSH (classical bound 2, quantum value 2.828427125), the chain with 4 settings (classical bound 6), I3322 (classical bound 0, upper bound 0.2509 at NPA level 2+AAB) and tilted CHSH with formula $\sqrt{8+2p^2}$ [C-selbsttest].

4 Results

Proposition A (computed_rigorous). For $p = 1/2$, $Q = 3.13693731$ up to $1e-6$ [C-bell-R1]. The verifier certified $3.13693731130907 \leq Q \leq 3.13693732592937$ (gap $1.46e-08$; strategy $d=2$, NPA 1+AB) [C-bell-R1, C-bell-R3]. The level 2+AAB upper bound is 3.13693731768845 [C-bell-R3-RT2]. An independent see-saw strategy gave the exact rational Bell value 3.13693731130907 with $d=2$ [C-bell-R1-RT1].

Proposition B (computed_rigorous). For $p = 0.3$, $Q = 2.93828832$ up to $1e-6$, with $2.93828833175884 \leq Q \leq 2.938288368081$ (gap $3.63e-08$; strategy $d=2$, NPA 1+AB) [C-bell-R4]. An independent $d=2$ strategy gave the exact value 2.93828833175884 [C-bell-R4-RT2].

Proposition C (computed_rigorous). For $p = 131/200$, $Q = 3.35825951388$ up to $1e-6$, with $3.35825951388355 \leq Q \leq 3.3582595532047$ (gap $4.14e-08$; strategy $d=2$, NPA 2) [C-bell-R6]. An independent $d=2$ strategy gave the exact value 3.35825951388355 [C-bell-R6-RT2].

Proposition D (computed_rigorous). The classical bound is exceeded at two parameter values.

- At $p = 0.99$, the verifier's explicit strategy has exact value 3.98000132884628 , and $L = 199/50$ [C-bell-R2]. The level 2+AAB upper bound is 3.98000623486376 [C-bell-R2-RT1].
- At $p = 0.95$, $L = 3.9$ [C-bell-R2-RT2]. Also $3.90016389936512 \leq Q \leq 3.90019981472908$ (gap $3.59e-05$) [C-bell-R2-RT2]. The gap is too large for certification of Q , but the lower bound exceeds L [C-bell-R2-RT2].
- At $p = 1$, $L = 4$ and the verifier's quantum strategy attains exactly 4, so no violation was found there [C-bell-R1-RT2].

Proposition E (computed_rigorous). At $p = 131/200$, $Q \leq 3.3582595532047$ (NPA level 2, dual certificate), while the optimum of NPA level 1+AB is ≥ 3.35826280973 (rational feasible moment matrix) [C-kombi1]. Level 1+AB is therefore strictly not tight there: its optimum exceeds Q by at least $3.25e-06$ [C-kombi1].

Proposition F (computed_rigorous). Lower bounds on (NPA 1+AB optimum) minus (best exactly certified strategy) are as follows [C-kombi2]:

p	lower bound on gap	source
0.5	-1.726e-08	[C-kombi2]
0.62	-2.336e-08	[C-kombi2]

p	lower bound on gap	source
0.652	1.843e-07	[C-kombi2]
0.6532	1.025e-06	[C-kombi2]
0.6533	1.120e-06	[C-kombi2]
0.655	3.296e-06	[C-kombi2]

Up to $p = 0.62$ the bound is below $1e-7$ in magnitude, which is the accuracy of the SDP solvers [C-kombi2]. From $p = 0.652$ on it is positive and grows monotonically over the points examined [C-kombi2]. A threshold defined by a fixed tolerance such as $1e-6$ therefore depends on that tolerance [C-kombi2]. At $p = 0.62$, the strategy value 3.30352972153 ($d \leq 3$) and the 1+AB value ≥ 3.30352969818 agree to within solver accuracy [C-bell-R5-RT1]. At $p = 0.72$, NPA level 2 reaches ≥ 3.46649094809 against a best strategy of 3.46649100491 ($d \leq 2$) [C-bell-R5-RT2].

Summary of certified values.

p	Lower bound (strategy)	Upper bound (NPA)	Gap	Status	Source
0.3	2.93828833175884 ($d=2$)	2.938288368081 (1+AB)	3.63e-08	proved	[C-bell-R4]
1/2	3.13693731130907 ($d=2$)	3.13693732592937 (1+AB)	1.46e-08	proved	[C-bell-R1]
131/200	3.35825951388355 ($d=2$)	3.3582595532047 (2)	4.14e-08	proved	[C-bell-R6]
0.95	3.90016389936512 ($d=2$)	3.90019981472908 (2+AAB)	3.59e-05	not certified	[C-bell-R2-RT2]
0.99	3.98000132884628	3.98000623486376 (2+AAB)	not stated	not certified	[C-bell-R2, C-bell-R2-RT1]

5 Negative results, preregistration and red team

Negative result. The scan for the transition at which NPA 1+AB ceases to be tight, over the points $p = 0.6, 0.63, 0.65, 0.655, 0.66, 0.7, 0.8, 0.9$ and 0.99 , produced no claim that passed verification [C-neg1].

Preregistration. H6a (at least 2 of 3 anchor questions confirmed two-sided) was not met: the integrator chose 0 of the 3 anchor questions, and 0 confirmed statements concern anchor questions [C-H6a]. H6b (at least one certified statement about an inequality outside the named families) was met, with 6 confirmed statements concerning the doubly tilted CHSH family or other unnamed inequalities [C-H6b]. Novelty relative to the literature is not claimed [C-H6b].

Red team. Red-team checks targeting the confirmed values failed in the sense that the attacks did not succeed.

- No strategy exceeded the certified values at $p = 1/2$, $p = 0.3$ and $p = 131/200$ [C-bell-R1-RT1, C-bell-R4-RT2, C-bell-R6, C-bell-R6-RT2].
- The upper bounds were not undercut by any strategy [C-bell-R3-RT2, C-bell-R2-RT1].
- No violation of L was found at $p = 1$ [C-bell-R1-RT2].
- NPA 1+AB was not found loose at $p = 1/2$ or $p = 0.62$, and NPA level 2 was not found loose at $p = 131/200$ or $p = 0.72$ [C-bell-R4-RT1, C-bell-R5-RT1, C-bell-R6, C-bell-R6-RT1, C-bell-R5-RT2].

One claim is contested [C-bell-R8]. At $p = 0.6533$, the 1+AB optimum lies at least $5e-07$ above the best strategy found by the verifier in dimension up to 4 [C-bell-R8]. Here the verifier reports 3.35554254508 against 3.35554142523, a gap $\geq 1.120e-06$ [C-bell-R8]. This does not show that Q itself lies below the 1+AB value [C-bell-R8]. A red-team check at $p = 0.6532$ passed (statement contested), with a gap $\geq 1.025e-06$ [C-bell-R8-RT1]. A

check at $p = 0.652$ did not show a gap of $1e-6$, with a gap $\geq 1.843e-07$ [C-bell-R8-RT2]. The agent's unverified interval for the transition is not used here.

6 Limitations and open questions

- **Not shown:** a closed formula for $Q(p)$. The fit attempts addressed this question [C-bell-R3, C-bell-R4], but no certified formula resulted.
- **Not shown:** that $Q > L$ for all $p < 1$, or that $Q = L$ at $p = 1$. The claims certify a strategy value of 4 at $p = 1$ and violations only at the tested points [C-bell-R1-RT2, C-bell-R2].
- **Not shown:** that qubits suffice in general. Certified strategies at the tested points have $d=2$ or $d \leq 3$, and the lower bounds there match the upper bounds only to the stated gaps [C-bell-R5-RT1, C-bell-R6].
- A sharp transition value for NPA 1+AB is not established. The verifier bounds are tolerance-dependent [C-kombi2], and the contested claim [C-bell-R8] is not resolved.
- Q is not certified at $p = 0.95$ and $p = 0.99$ [C-bell-R2-RT2, C-bell-R2-RT1].
- No novelty is claimed beyond what the claims support [C-H6b].

Related work. The NPA hierarchy yields exact bounds via an infinite series of semidefinite programs [C-lit6]. In the simplest Bell scenario, the correlations at levels 1+AB and 2 coincide under a plausible criterion [C-lit13]. Analytic criteria for the boundary of quantum correlations are hard to establish even there [C-lit14]. An iterative algorithm computes the quantum maximum of two-outcome Bell inequalities in a given Hilbert space dimension [C-lit10]. Operations on CHSH-type inequalities that preserve the Tsirelson bound but change the classical bound have been studied [C-lit5]. For I3322, finite-dimensional systems are conjectured not to attain the true quantum maximum [C-lit9]. Proposition E shows that, for a functional with marginal terms in the simplest scenario, the optima of NPA levels 1+AB and 2 differ [C-kombi1]. The coincidence of levels 1+AB and 2 reported in [C-lit13] is stated for correlations under a plausible criterion; whether its scope covers functionals with marginal terms was not checked beyond the abstract, so we do not claim a contradiction. Beyond this, we make no claim about how these results relate to the present family.

Verification log (automatic): 34 claim IDs cited; unsupported statements per writing round: $30 \rightarrow 2$; 0 sentences removed; remaining violations: 0. Claim texts and evidence: `paper_belege.json`.